

REVIEW 1 · PROJECT PRESENTATION

# AR-EAURP

Adversarial-Resilient & Energy-Harvesting Aware EAURP

*Securing DRL-Based Trust Routing in MANETs Against Trust Poisoning and  
Enabling Dynamic Renewable-Energy Awareness*

---

**Base Paper:** DRL-EAURP — Lekha S., VIT Vellore

**Extension Roadmap:** AR-EAURP (24-07-26)

2 August 2026

# What We'll Cover Today

**A**

Limitations of Existing Works

**B**

Advancement of the Base Paper (DRL-EAURP)

**C**

How the Base Paper Solves the Problem

**D**

Limitations of the Base Paper

**E**

Senior's Proposed Approach — AR-EAURP

**F**

Our Critique of the Senior's Approach

**G**

Proposed Extensions — Work Plan

**H**

Timeline to Review 3 (10 Weeks)

# Why Routing Is Hard in MANETs

*Mobile Ad Hoc Networks have no fixed infrastructure — every node is both a host and a router.*

1

## MOBILITY

Nodes move continuously; links form and break, making static routes unreliable.

2

## ENERGY

Every node has finite battery; poor route choices drain nodes early and fragment the network.

3

## TRUST

Not all nodes forward packets honestly — misbehaving or malicious nodes degrade delivery.

*Shortest-path routing alone cannot guarantee reliable communication — routing must jointly reason about mobility, energy, and trust.*

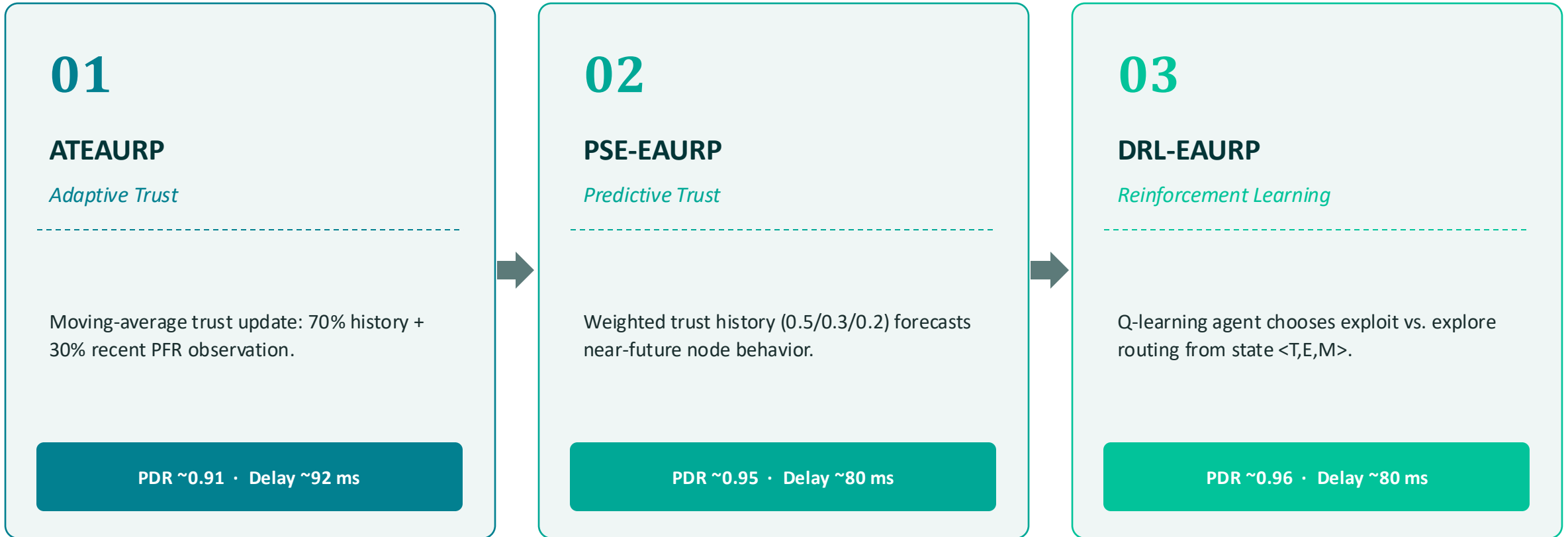
# Limitations of Existing Routing Approaches

|                              |                                                   |                                                                    |
|------------------------------|---------------------------------------------------|--------------------------------------------------------------------|
| <b>Trust-only</b>            | <i>T2AR, Trust-Based Secure Routing</i>           | Ignore node energy — reliable nodes get over-used and die early    |
| <b>Energy-only</b>           | <i>ECC-Based, Secure Energy-Efficient Routing</i> | No trust evaluation — vulnerable to malicious / misbehaving nodes  |
| <b>Privacy-only</b>          | <i>ANODR, Trust-Aware Anonymous Routing</i>       | Strong anonymity but heavy overhead; energy left unoptimized       |
| <b>Hybrid (Trust+Energy)</b> | <i>ETERS, EAURP, QoS-Aware Trust Routing</i>      | Combine trust & energy but stay static — no adaptive learning      |
| <b>AI / Learning-based</b>   | <i>IRDFPR-CMDNN, GNN-based, Swarm Routing</i>     | Learn intelligent policies but carry high computational complexity |

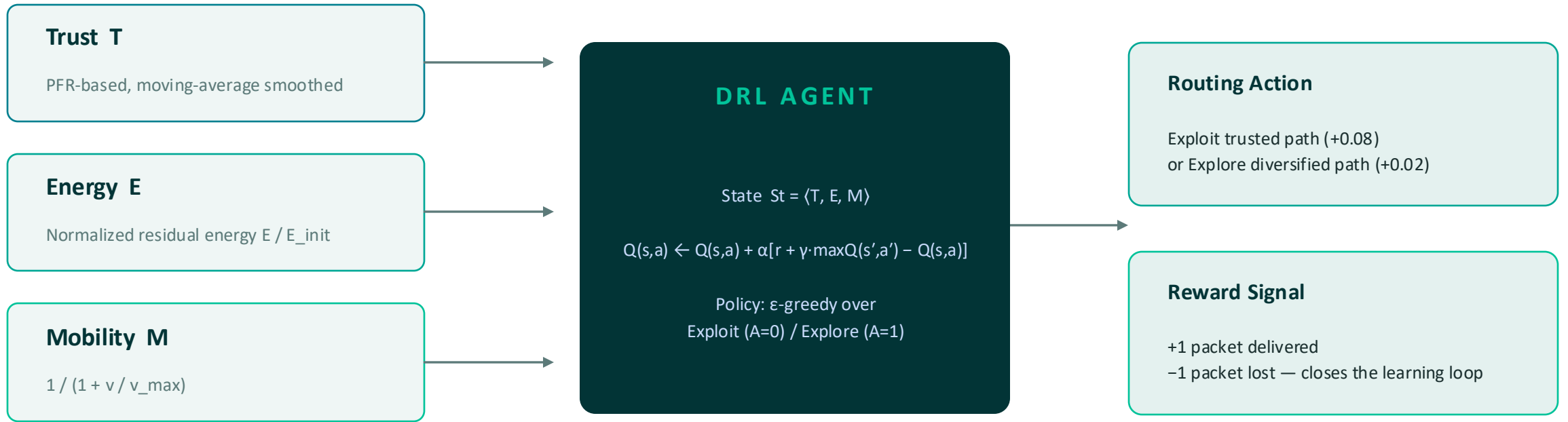
Very few protocols unify trust, energy, and adaptive intelligence at once — the gap the base paper targets.

# DRL-EAURP: A Three-Stage Evolution

*Each stage progressively improves on the last, moving from reactive trust to a fully learning-based routing agent.*



# How DRL-EAURP Makes a Routing Decision



*The agent continuously balances reliability, energy, and mobility — instead of fixing weights by hand, it learns them from network feedback.*

# DRL-EAURP Outperforms All Prior Variants

0.96

Peak Packet  
Delivery Ratio

~80 ms

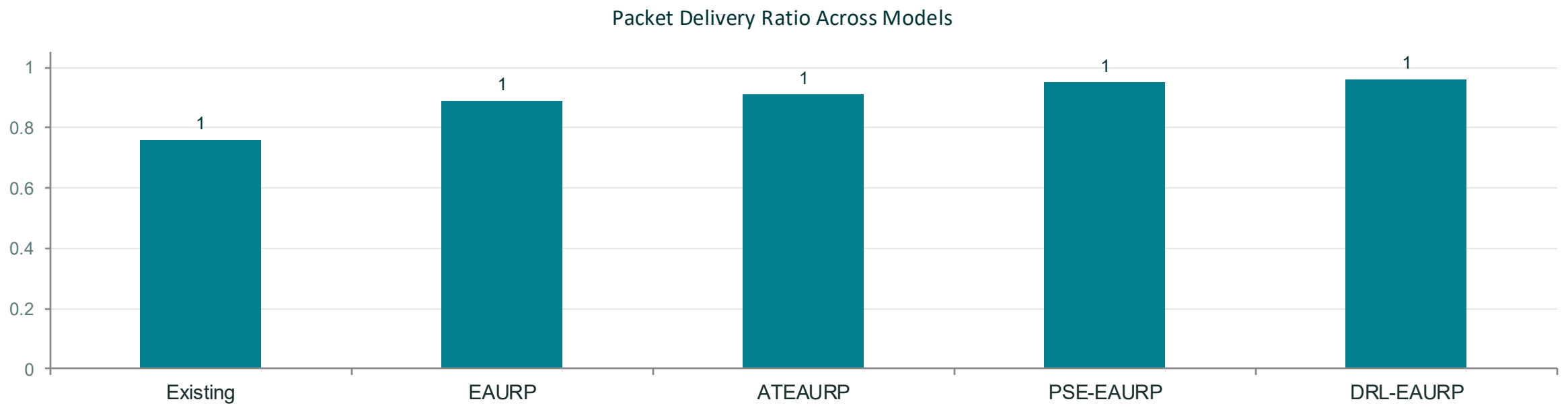
Average  
Delay

5.8 kbps

Stable  
Throughput

~1230

Rounds of  
Network Lifetime



# Limitations of DRL-EAURP

## PFR-based trust is gameable

A node dropping only ~15% of packets stays above the 0.6 trust threshold — a gray-hole / low-and-slow attack goes undetected.

## Static, linear energy model

Eq. (8) assumes fixed depletion only; it cannot represent renewable sources such as solar or kinetic harvesting.

## Binary action space

Exploit / Explore ( $A_t \in \{0,1\}$ ) is coarse — it cannot express graded routing strategies.

## Simulation-only validation

Results come from a Python/Colab simulator, not a network-realistic testbed (e.g., NS-3) or hardware trial.

## No collusion / slander defense

Trust is computed per-node in isolation; malicious nodes falsely vouching for each other are not modeled.



# AR-EAURP: The Senior's Solution

*Adversarial-Resilient & Energy-Harvesting Aware EAURP — directly targets the two gaps identified above.*

## CORE ISSUE

**Trust poisoning:** gray-hole / low-and-slow attackers stay under the 0.6 trust threshold while still crippling PDR.

**Static energy assumption:** the DRL policy has no way to plan around nodes with renewable (solar) energy income.

1

## GAN-DRL — Robust Adversarial Training

A simulated adversary actively manipulates trust metrics during training, so the DRL policy learns routing that stays robust under attack.

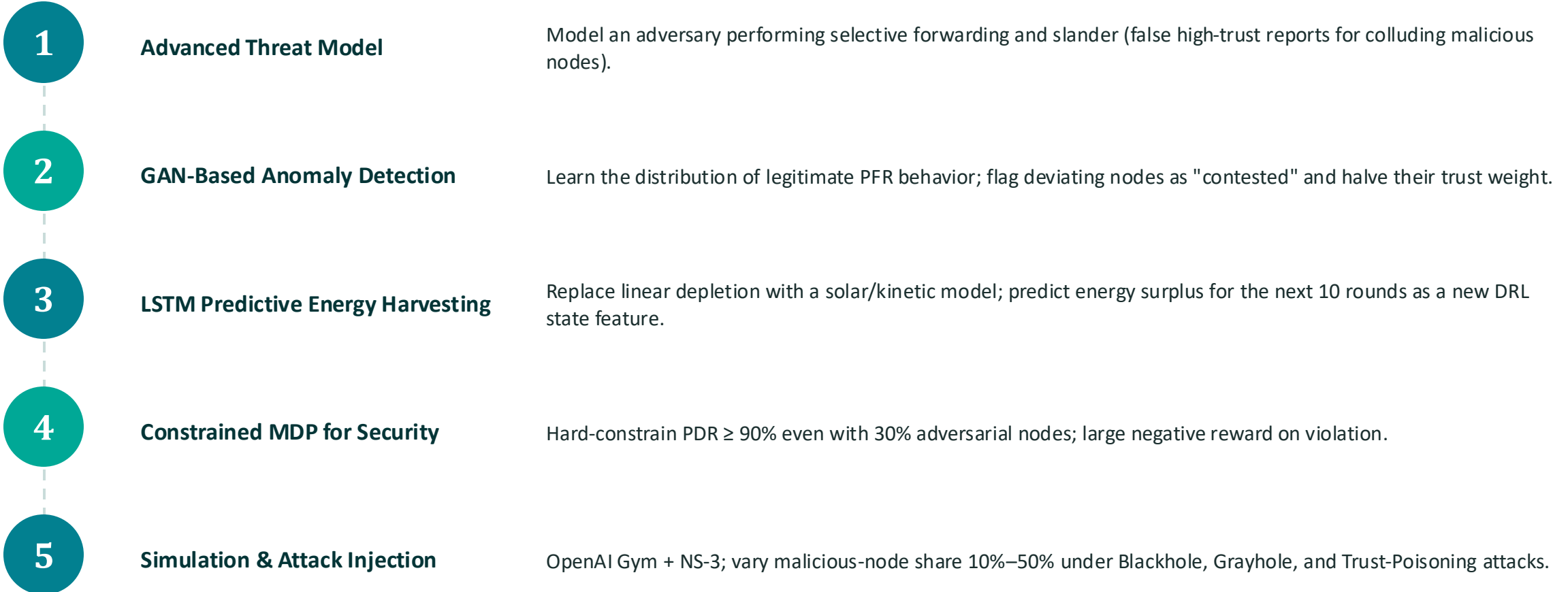
2

## LSTM — Hybrid Energy Prediction

Forecasts solar / kinetic energy income for the next 10 rounds, feeding "predicted energy surplus" into the DRL state.

*Target: sustain PDR above 90% even when adversaries control up to 40% of network nodes.*

# Five-Step Plan Proposed by the Senior



# Open Gaps in the AR-EAURP Roadmap

## GAN training is non-trivial

Mode collapse and the need for large, labeled attack traces make convergence uncertain on limited simulation data.

## Energy model needs real data

LSTM-based harvesting prediction is only as good as the solar/kinetic traces it is trained on — none are specified yet.

## CMDP feasibility risk

A hard 90% PDR floor under sustained attack may be infeasible, causing reward-shaping oscillation instead of a stable policy.

## Compute overhead on-device

GAN + LSTM + DRL together may be too heavy for resource-constrained MANET nodes — no lightweight/edge variant is planned.

## Narrow attack coverage

Focuses on gray-hole, slander, and trust poisoning; Sybil and wormhole attacks — common in MANETs — are not addressed.

# How We Plan to Extend the Work

## Lightweight anomaly module

Replace the full GAN with a compact autoencoder for near-edge deployment on constrained nodes.

## Federated trust learning

Share trust signals across nodes without exposing raw traffic, improving detection of colluding attackers.

## Sybil & wormhole detection

Add a lightweight identity/topology check layer alongside the GAN anomaly detector.

## Real energy-trace validation

Use public solar-irradiance datasets to train/validate the LSTM harvesting model.

## NS-3 realistic PHY layer

Move from a Python/Colab simulator to NS-3 for MAC/PHY-accurate delay and loss.

## Explainable DRL policy

Log Q-value attributions so routing decisions can be audited, not just observed.

# Work Plan — 10 Weeks to Review 3

Review 3 is due in 2.5 months. Phases follow the senior's roadmap, sequenced with buffer for consolidation.



**Milestone:** Review 3 evaluation package ready — end of Week 10

# Key Takeaways

1

Existing MANET routing work solves trust, energy, privacy, or intelligence in isolation — rarely together.

2

DRL-EAURP unifies all four through a 3-stage trust → prediction → reinforcement-learning pipeline, reaching 0.96 PDR.

3

Its PFR-based trust and static energy model remain exploitable and unrealistic under adversarial/renewable conditions.

4

AR-EAURP (senior's plan) targets exactly these gaps with GAN-based adversarial training and LSTM energy prediction.

5

Our work plan adds lightweight, federated, and explainable extensions — validated on a 10-week path to Review 3.

# Thank You

Questions & Discussion